Faculty & Department Effectiveness (FEI)

Student Early Alert System — metrics, scoring formulas, and plain-language guide.
Version: June 2026 · Matches src/lib/effectiveness.ts

Purpose: Measure whether faculties use the alert system to support student wellbeing — not only to flag at-risk students.

1. What is FEI?

FEI (Faculty Effectiveness Index) is a score from 0 to 100 with letter grade A–E. It combines outreach, wellbeing referrals, student recovery, and data quality.

In simple terms: When the system flags a struggling student, does this faculty help them — and do students improve?

Grade	FEI	Meaning
A	85–100	Exemplary support and outcomes
B	70–84	Effective with minor gaps
C	55–69	Developing — weak response or outcomes
D	40–54	At risk — major support gaps
E	<40	Critical — support largely absent

2. Overall FEI formula

```
FEI = round( 0.30×Outcome + 0.25×Wellbeing + 0.25×Response + 0.10×Readiness +
0.10×Sustained , 2 )
```

Component	Weight	Measures
Outcome	30%	Recovery + repeat-alert control
Wellbeing	25%	Referrals + wellbeing uptake
Response	25%	Coverage, speed, stale cases
Readiness	10%	Attendance posting quality
Sustained	10%	Recovery + repeat (emphasis)

3. Percentage helper

```
If denominator d > 0:  Rate% = round( (n ÷ d) × 100 , 2 decimals )
If d = 0:  Rate% is empty (shown as —)
```

4. Raw counts

Count	Meaning
N_total	Students enrolled in faculty/department
N_alerted	Students with warning or critical alert
N_crit_alerted	Students with critical alert
N_intervened	Alerted students with "e1 intervention
N_crit_intervened	Critical alerted + intervention
N_referred	Latest intervention status = referred
N_wellbeing	Referred + wellbeing case exists
N_recovered	Intervened + no longer in alert
N_repeat	In alert but prior case was resolved
N_open / N_stale	Open cases / open >14 days

5. Dashboard columns

Coverage %

```
Coverage% = pct(N_intervened , N_alerted)
```

Plain English: Of flagged at-risk students, what % did staff contact? Example: 74÷165 = 44.9%

Critical coverage %

```
Critical coverage% = pct(N_crit_intervened , N_crit_alerted)
```

Plain English: Same as coverage, but only for critical (most serious) alerts.

TTFC (median days)

```
″_S = (first intervention date " first alert date)
TTFC = median(″_S)
```

Plain English: How fast staff respond after the first alert. Lower is better. — if history missing.

Wellbeing %

```
Wellbeing% = pct(N_wellbeing , N_referred)
```

Plain English: Of referred students, how many reached the wellbeing centre records?

Recovery %

```
Recovery% = pct(N_recovered , N_intervened)
```

Plain English: After intervention, how many students cleared their alert?

Alerted

```
Alerted = N_alerted
```

Plain English: Current number of at-risk students — workload indicator, not quality alone.

6. Scoring each parameter (0–100)

Each rate is mapped to a score using bands. Missing data !' score 50 (neutral).
Higher is better — f!'(x)

Condition	Score
x "e Excellent	100
x "e Good	85
x "e Fair	70
x "e Poor	55
Below poor	35
Missing	50

Lower is better — f!“(x)

Condition	Score
x "d Excellent	100
x "d Good	85
x "d Fair	70
x "d Poor	55
Above poor	35
Missing	50

Band thresholds

Parameter	Type	Exc.	Good	Fair	Poor
Coverage %	!‘	"e 95	"e 85	"e 70	"e 50
Critical cov. %	!‘	"e 95	"e 90	"e 75	"e 60
TTFC (days)	!“	"d 3	"d 7	"d 14	"d 21
Stale %	!“	"d 10	"d 20	"d 35	"d 50
Referral rate %	!‘	"e 40	"e 25	"e 15	"e 8
Wellbeing %	!‘	"e 80	"e 65	"e 50	"e 35
Recovery %	!‘	"e 60	"e 45	"e 30	"e 15
Repeat alert %	!“	"d 5	"d 10	"d 20	"d 30
Attendance post. %	!‘	"e 95	"e 90	"e 80	"e 70

7. Component scores

Response (25% of FEI)

```
Response = avg( f!‘(Coverage), f!‘(Critical cov.), f!
```

Plain English: Reaching flagged students quickly, including critical cases, without abandoning open cases.

Wellbeing (25% of FEI)

```
Wellbeing = avg( f!‘(Referral rate%), f!‘(Wellbeing
```

Plain English: Identifying distress and handing students to wellbeing services.

Outcome (30% of FEI)

```
Outcome = avg( f!‘(Recovery%), f!“(Repeat alert%) )
```

Plain English: Students improving and not cycling back into crisis.

Readiness (10% of FEI)

```
Readiness = f!‘(Attendance posting%)
```

Plain English: Reliable attendance data so alerts are trustworthy.

Sustained (10% of FEI)

```
Sustained = avg( f!‘(Recovery%), f!“(Repeat alert%)
```

Plain English: Same as outcome — emphasises lasting improvement.

8. Worked example — Social Sciences

Step	Calculation	Result
Coverage	74 ÷ 165 × 100	44.9%
Score	44.9% < 50 poor band	35
Recovery	0 ÷ 74	0% !’ score 35
TTFC	No daily history	— !’ 50
FEI	Weighted blend	"H57 (C)

Despite best coverage in the university, FEI stays at C because recovery is 0%, wellbeing handoff is not visible, and TTFC cannot be measured.

9. Key reminders

- FEI highlights support gaps — not student failure.
- High Alerted often means detection works.
- Low coverage + high alerts = outreach has not kept pace.
- Scores refresh daily via ETL (/api/cron/effectiveness).